Choose your own maths adventure

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The Room (Fireproof Games)

1. What did we do?

(1) — (2) — (3) — **(4)** — (5) — (6) — (end)

Part 4

Consider

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

Question: Find the projection of $\mathbf{b}$ onto the columnspace of A by solving $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$ and $\mathbf{p} = A\hat{\mathbf{x}}$.

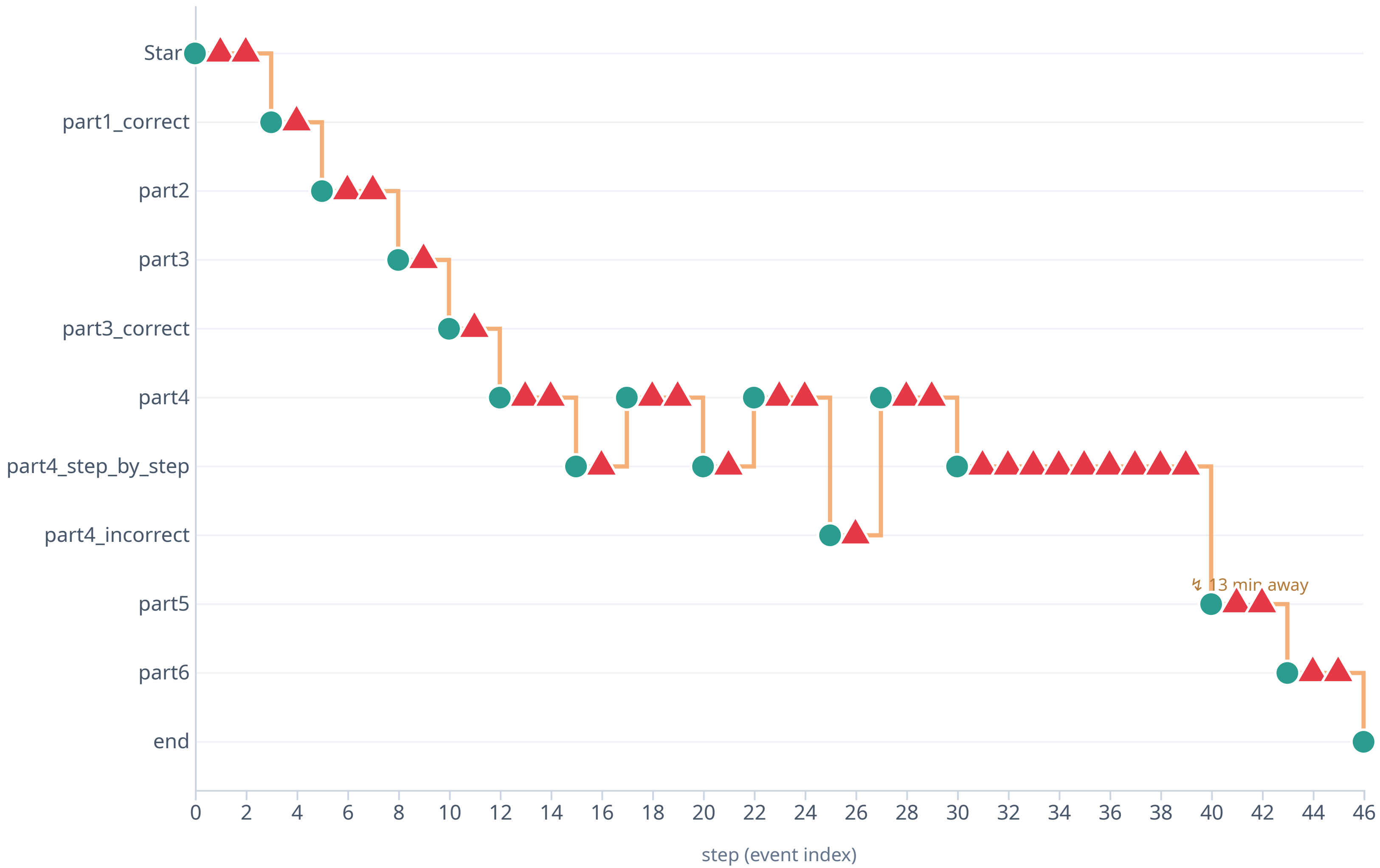
Reveal hint

Show me step-by-step

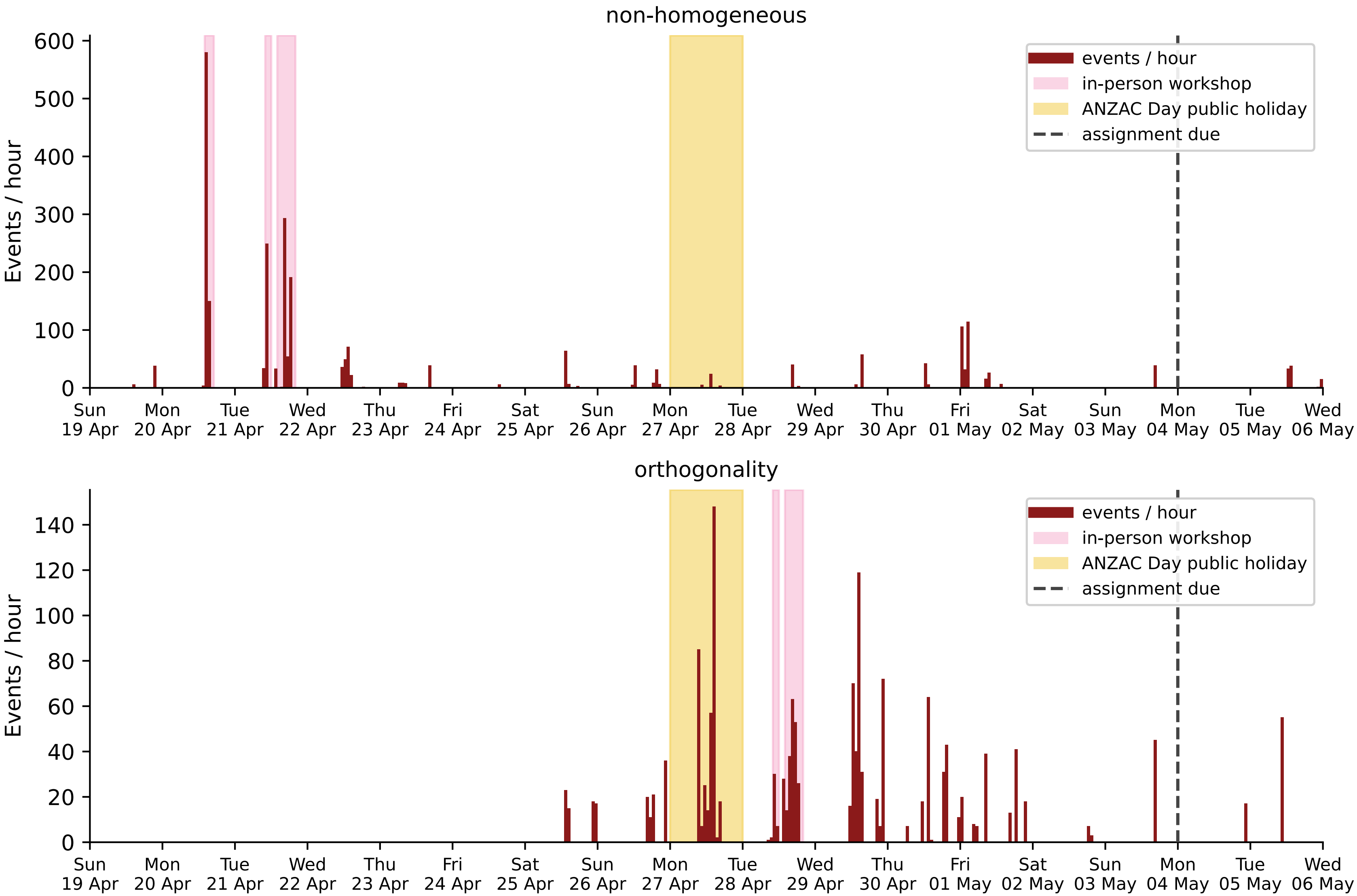
I'm ready to answer

1. What did we do?

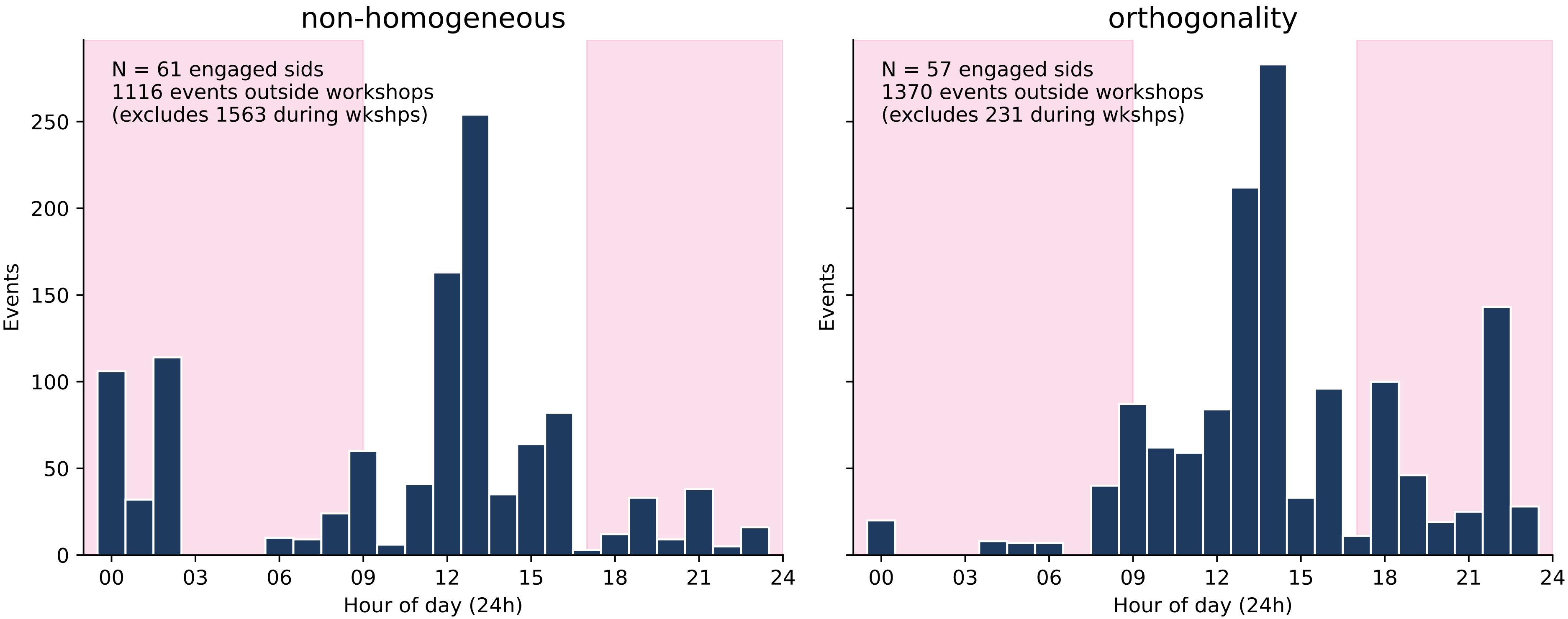
One student's journey



2. Who did we reach?



2. Who did we reach?

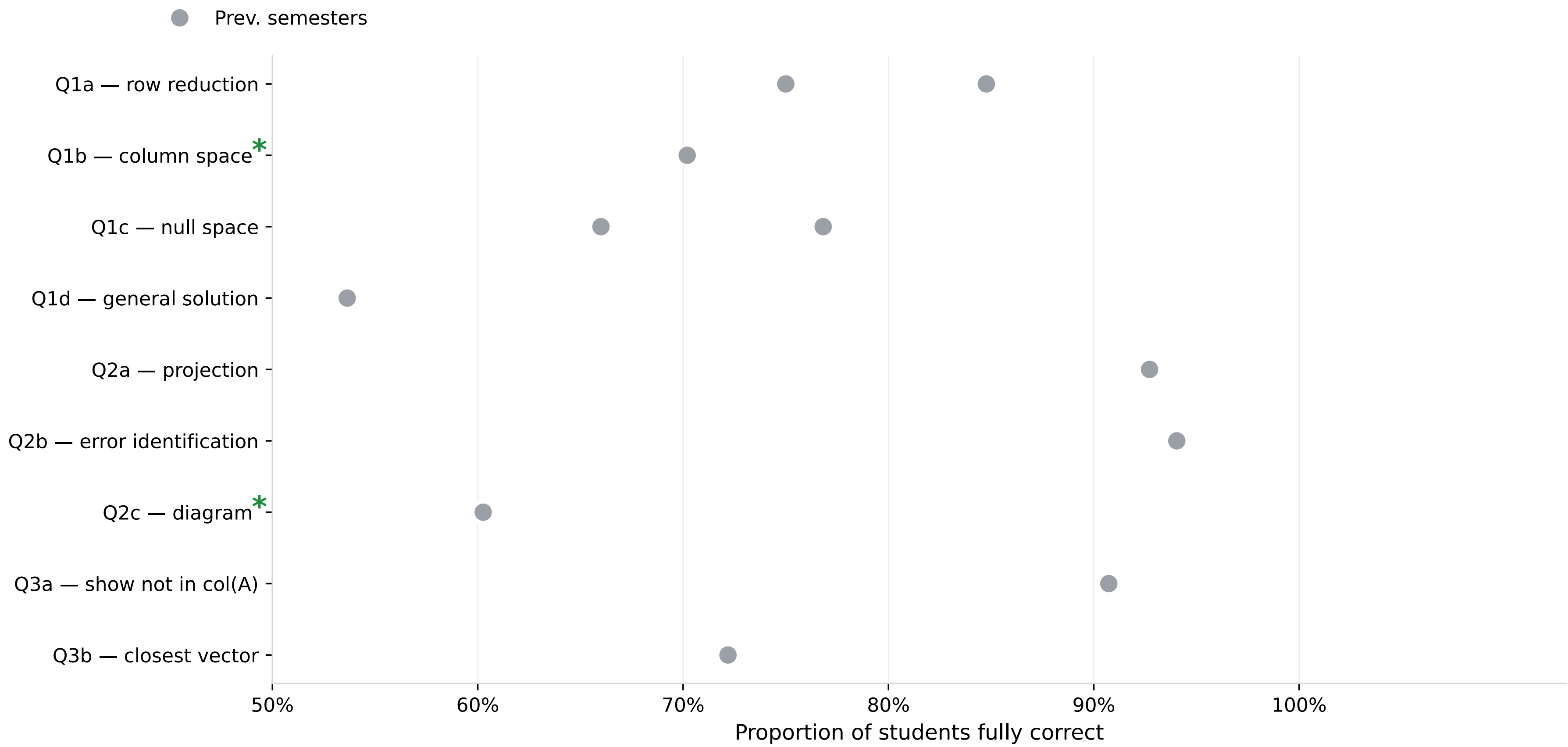


3. Did it work?

> *"I was just emailing you to let you know that I have found the workshop hints to be super informative, intuitive and helpful. It would be awesome if the teaching staff could produce them on a weekly basis."*
>
> — student email to Unit Coordinator

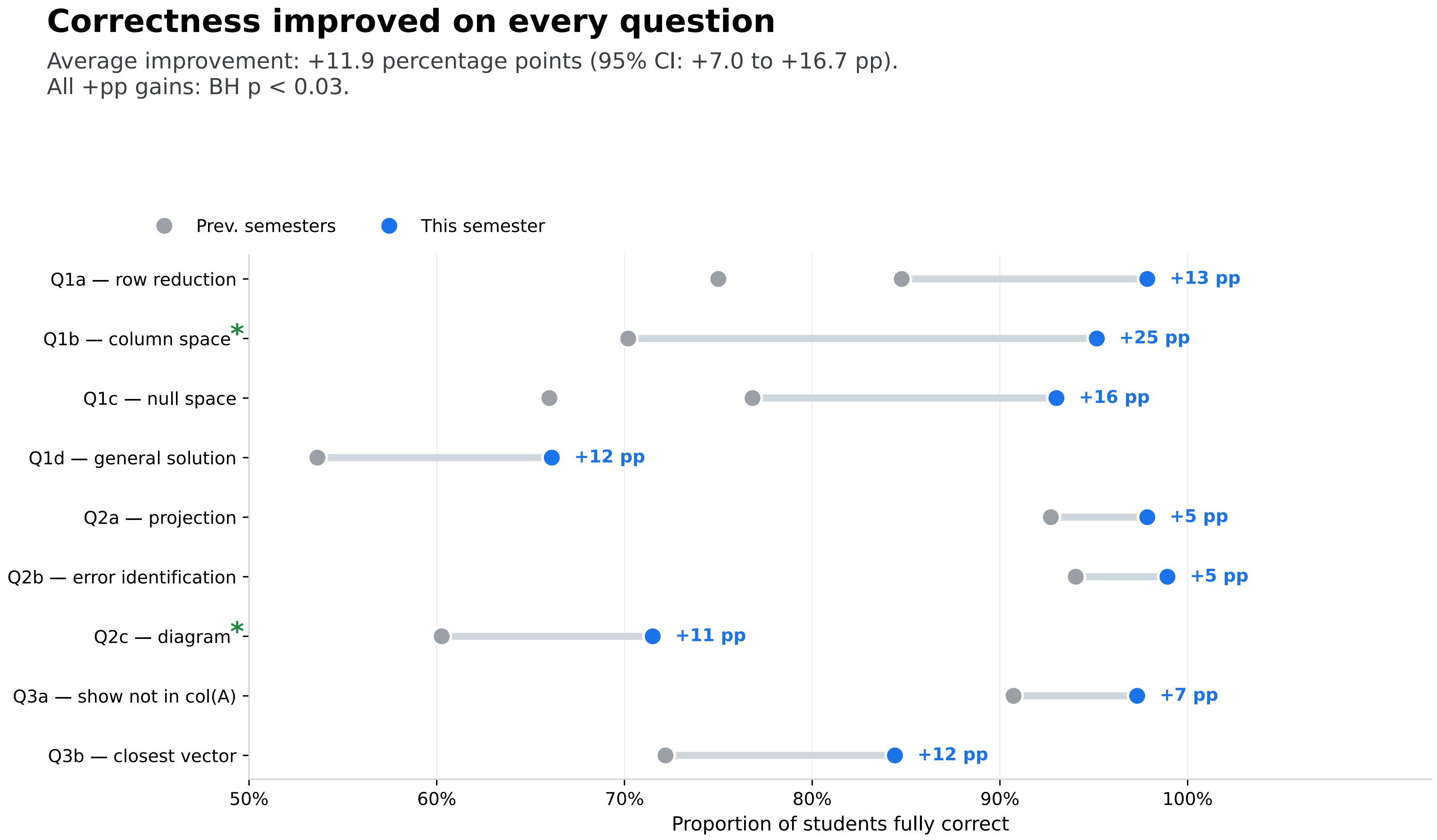
"It read my mind" *"Better than Khan Academy"*

3. Did it work?



Prev. semesters
Q1a — row reduction
Q1b — column space*
Q1c — null space
Q1d — general solution
Q2a — projection
Q2b — error identification
Q2c — diagram*
Q3a — show not in col(A)
Q3b — closest vector
50%
60%
70%
80%
90%
100%
Proportion of students fully correct

3. Did it work?



Correctness improved on every question
Average improvement: +11.9 percentage points (95% CI: +7.0 to +16.7 pp).
All +pp gains: BH p < 0.03.
Prev. semesters
This semester
Q1a — row reduction
Q1b — column space*
Q1c — null space
Q1d — general solution
Q2a — projection
Q2b — error identification
Q2c — diagram*
Q3a — show not in col(A)
Q3b — closest vector
+13 pp
+25 pp
+16 pp
+12 pp
+5 pp
+5 pp
+11 pp
+7 pp
+12 pp
50%
60%
70%
80%
90%
100%
Proportion of students fully correct

3. Did it work?

Correctness improved on every question

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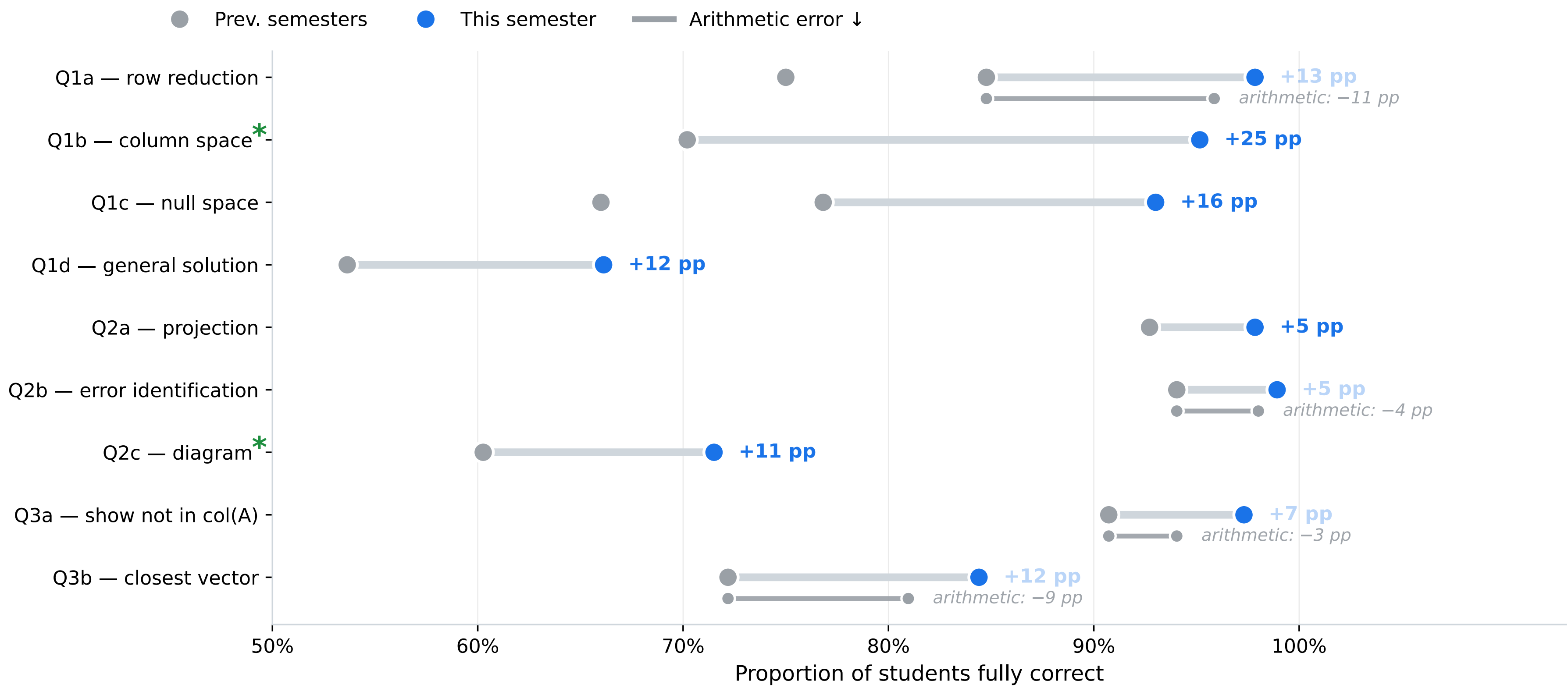

3. Did it work?

Correctness improved on every question

Average improvement: +11.9 percentage points (95% CI: +7.0 to +16.7 pp).
All +pp gains: BH $p < 0.03$.
Sub-bars show the dominant mechanism behind each question's gain * BH $p < 0.05$, ** BH $p < 0.01$.
Both 'missing explanation' effects also had significant 'concept right' gain (excludes arithmetic errors)